



Building ML-Based Compliance Monitoring for Automated Risk Detection

A practical case study of deploying a Machine Learning-powered **Working Hours Alert System (WHAS)** — automating compliance monitoring, detecting time-related fraud risks, and enabling human-led investigation at enterprise scale.

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The Compliance Monitoring Problem



Why Manual Processes Break Down

As organizations scale, compliance data — attendance, scheduling, leave, and policies — fragments across systems. Reviewers spend more time **collecting data** than investigating risk.

The Hidden Gap

Manual processes focus on *known* problems. Emerging behavioral patterns go unnoticed until they trigger an existing control — or don't.

From Rule Detection to Risk Detection

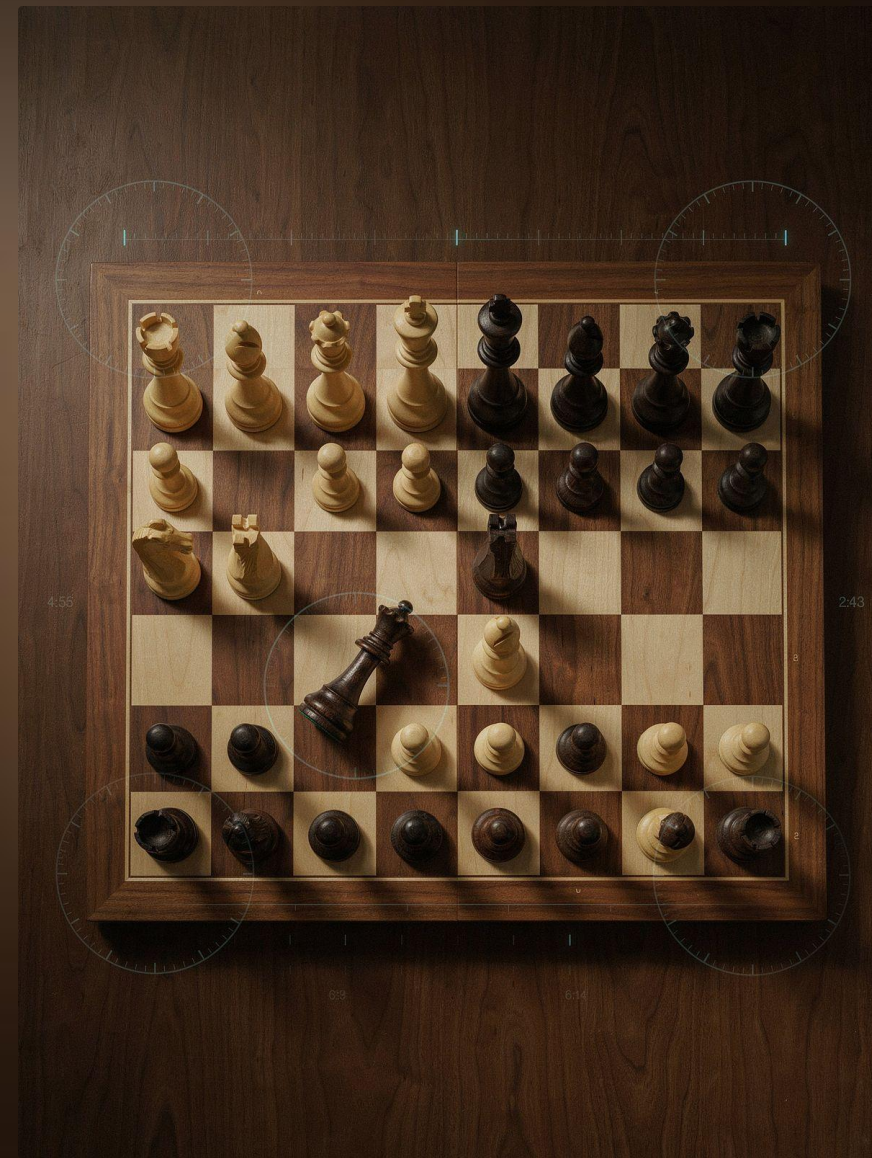
Static rules catch **defined violations**. But risk often hides in subtle patterns — repeated timing anomalies, irregular durations, deviations from an individual's baseline — that no single rule can capture.

Rules Engine

Deterministic. Catches known, explicit violations.

ML Anomaly Detection

Adaptive. Surfaces unusual patterns across multiple signals simultaneously.



System Design: WHAS Architecture

WHAS is a **decision-support system**, not an enforcement mechanism. Anomalies are investigative leads — not conclusions.



Each stage has a distinct responsibility. Humans retain authority over interpretation and action.

Building the Data Foundation

Data arrives from attendance, scheduling, leave, and HR systems — each representing time differently. Inconsistencies and duplicates are validated and resolved *before* data reaches the model.

📌 Data quality often matters more than model complexity.

Ingest

Attendance, scheduling, leave & HR

Normalize

Standardize formats, align
timestamps

Validate

Flag inconsistencies before
modeling

Feature Engineering for Behavioral Analysis



Duration & Timing

Working duration, timing consistency, and deviation from expected schedules.



Recurring Patterns

Repeated activity patterns and gaps between events over time.



Historical Baseline

Changes relative to the individual's own historical behavior.



Context

Shift type, approved leave, travel, and operational circumstances shape what "normal" means.

Choosing the Right ML Approach

Compliance datasets have many examples of **ordinary activity** but few confirmed violations — making supervised classification difficult. Anomaly detection learns expected patterns and flags deviations.

A **hybrid architecture** works best: deterministic rules for explicit policy requirements, ML for subtle behavioral deviations.

- ❏ Interpretability beats sophistication. Reviewers must understand *why* an alert fired.



A magnifying glass with a gold rim and a dark wooden handle is positioned over a document on a light-colored wooden surface. The document features a bar chart with orange bars and a line chart with a blue line. The bar chart has a y-axis ranging from 0 to 1000 and an x-axis with labels 2015, 2016, 2017, 2018, 2019, and 2020. The line chart has a y-axis ranging from 0 to 2500 and an x-axis with labels 2015, 2016, 2017, 2018, 2019, and 2020. The magnifying glass is focused on the 2017 data point of the bar chart.

Anomaly Scores Are Not Conclusions

A risk score indicates that something **deserves attention** — not that misconduct occurred. Operational incidents, schedule changes, system errors, or incomplete records can all produce anomalous patterns.

What the Model Provides

Observed pattern, contributing signals, historical comparison, and contextual information.

What the Reviewer Decides

Whether the alert represents a legitimate explanation, data issue, policy exception, or need for further investigation.

Designing Actionable Alerts

Avoiding Alert Fatigue

Overly sensitive systems bury important signals in noise. Alert prioritization should weigh anomaly strength, recurrence, signal consistency, and available context.

Explainability by Design

Alerts must show *which characteristics* triggered detection — not just that a record is unusual. This enables focused, efficient investigation.



Human Oversight & Explainability

Human-in-the-Loop

Automation handles large-scale pattern detection. People retain authority over consequential interpretations and actions.

Feedback Loop

Reviewer outcomes reveal which signals are useful, enabling governed improvement of thresholds and features.

Explainability

Built in from day one — not added after deployment. Supports governance, debugging, and reviewer trust.



Enterprise Integration & Reliability

Secure Ingestion

Encrypted data transfer with access controls at every entry point.

Observability

Monitor infrastructure health, data freshness, feature validity, and model behavior continuously.

Graceful Degradation

When critical data is missing, mark analysis incomplete rather than generate a misleading alert.

Audit Trail

All pipeline changes, model versions, and alert logic must be fully traceable.

When critical data is missing, the safest response is marking analysis **incomplete** — not generating a misleading alert. Monitoring must cover infrastructure health, data freshness, feature validity, and model behavior.

Privacy, Security & Responsible Monitoring

Data Minimization

Collect only what has a defensible relationship to the monitoring objective. Each data element must be justified.

Governance Covers the System Too

Model versions, alert logic, configuration changes, and data transformations must all be traceable — preventing the monitoring platform itself from becoming a risk.



Managing False Alerts & Model Drift

1

Balance Sensitivity

Too sensitive → alert fatigue. Too lenient → missed risks. The right threshold is refined through operational experience, not set at launch.

2

Monitor for Drift

Workforce behavior evolves. Policies change. Teams shift. A model trained on historical data can become less representative over time.

3

Diagnose Before Retraining

Determine whether a shift reflects legitimate organizational change, a pipeline problem, or actual new risk behavior before triggering retraining.

Key Engineering Lessons

Start with the Workflow

Understand the compliance process before selecting an algorithm. ML should solve an operational problem, not showcase technical sophistication.

Transparency Drives Adoption

Explainable alerts help reviewers treat ML as an analytical assistant — not an unexplained authority.

Data Quality First

Reliable timestamps and clean records create more value than complex models. Reviewer trust depends on trustworthy data.

Automate Wisely

Machines process patterns at scale. Humans provide context, judgment, and accountability for consequential decisions.

The Future of Intelligent Compliance

The WHAS architecture extends to **any domain** where risk emerges through patterns: access monitoring, transaction analysis, process integrity, security operations.

The Governing Cycle

Observe → Evaluate → Review → Improve → Validate →
Deploy → Monitor

The Core Principle

Appropriate automation — not maximum automation.
Transparency, accountability, and human authority must remain intact.

Thank You

Questions, feedback, or collaboration opportunities are welcome.

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ML Engineering | Compliance Systems | Risk Detection